

Kushagra Gupta

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SUMMARY

Software Development Engineer with 3 years at Reliance JIO Platforms, engineering backend systems and real-time data pipelines that process **2 billion+** daily records at production scale. With hands-on expertise in Node.js, Python, TypeScript, Apache Kafka, Elasticsearch, Redis, and Apache Druid across ETL/ELT design, distributed systems, and observability tooling.

EXPERIENCE

Software Development Engineer - I

Dec 2023 – Present

Reliance JIO Platforms Limited

Mumbai, India

- Engineered core reporting infrastructure handling **2 billion+ daily records** by refactoring ingestion pipelines, optimizing memory bounds, and automating lifecycle archival, increasing throughput by **40%**.
- Owned the end-to-end analytics module, engineering data retrieval and aggregation pipelines across **Apache Druid** (OLAP), **MongoDB**, and **Elasticsearch** to surface business intelligence on SMPP traffic patterns, billing accuracy, and service observability.
- Designed **ETL/ELT** pipelines using **Confluent Kafka** sink/source connectors for MongoDB and Elasticsearch, enabling low-code auditable data flow with enhanced audit log searchability and long-term data persistence.
- Optimized **Redis** connection pooling (**78%** efficiency gain) and built an **Elasticsearch-Redis** bidirectional sync mechanism, slashing retrieval latency by **30%** with automated monitoring via **Prometheus** and **Grafana**.
- Contributed to the zero-downtime migration of 2 infrastructure clusters from **Azure Cloud** to **Jio Business Cloud** (in-house), improving scalability and operational reliability while reducing infrastructure costs.
- Led modernization of legacy services and operational portals through full version upgrades, leveraging AI-driven development (**Cursor**, **Claude**) to accelerate feature delivery and reduce deployment cycle time.
- Implemented IP whitelisting for telemarketer traffic to enforce trusted-source-only access, and engineered proactive alerting via **email** and **WhatsApp** across microservices for faster incident response and reduced MTTD.
- Engineered internal monitoring dashboards in **ReactJS** with reusable component architecture, enabling dynamic consumer control and reducing Mean Time to Recovery (**MTTR**) by **35%**.

DevOps Intern

May 2022 – Jul 2022

Celebal Technologies

Jaipur, India

- Automated deployment pipelines for static websites on **Azure** using **CI/CD** and **Docker**, reducing manual deployment effort by **70%**.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL

Frameworks & Libraries: Node.js, Express.js, FastAPI, React, Next.js, Angular

Databases & Streaming: MySQL, MongoDB, Redis, Apache Druid, Elasticsearch, Apache Kafka

Cloud, DevOps & Monitoring: Azure, Docker, NGINX, Linux, Git, Prometheus, Grafana, CI/CD

AI & Dev Tools: Cursor, Claude, GitHub Copilot

Core Concepts: REST APIs, Microservices, ETL/ELT Pipelines, Event-Driven Architecture, Distributed Systems, RBAC

Certifications: Microsoft Azure Fundamentals (AZ-900)

PROJECTS

Bus Charging Scheduler | *Python, Streamlit, Rule Engine, Algorithm Design* | [Live](#) | [GitHub](#)

- Production-quality scheduler for **20 electric buses** on a **540 km** corridor using a weighted rule engine with tunable soft constraints (individual fairness, operator fairness, network efficiency); fully data-driven via JSON with zero hardcoded constants – deployed live on **Streamlit Cloud** with 5 stress-test scenarios.

Unified Task & Support Ticket Tracker | *Node.js, React, TypeScript, Socket.IO, Prisma, JWT, Vitest* | [GitHub](#)

- Full-stack system with **RBAC** (Admin/Manager/Member), state-machine-enforced status transitions, authenticated real-time **Socket.IO** notifications (JWT room isolation, offline resiliency), 18+ REST endpoints via **Prisma ORM**, and a full backend test suite using **Vitest** and **Supertest**.

EDUCATION

Rajasthan Technical University

Jaipur, India

Bachelor of Technology in Computer Science and Engineering | **CGPA: 8.04 / 10**

Aug 2019 – Jul 2023